

BrightLearn exercise 2: SQL Aggregate Functions & Operators

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Q1 List all ~~distinct~~ distinct departments in the students table

Query:

```
SELECT DISTINCT department  
FROM students;
```

Output table

<u>department</u>
IT
HR
Finance

Q2 Get average age of students per department

Query:

```
SELECT department, AVG(age) AS avg_age  
FROM students
```

```
Group BY department;
```

Output Table

<u>department</u>	<u>avg_age</u>
IT	20.5
HR	22
Finance	23

Q3 Show departments with more than 1 student

Query:

```
SELECT department,  
COUNT(*) AS student-count  
FROM student  
GROUP BY department  
HAVING COUNT(*) > 1;
```

Output table

department	Student Count
IT	2
HR	2

Q4 Get all students whose age is between 21 & 23

Query:

```
SELECT student id, name, age, department  
FROM students  
WHERE age  
BETWEEN 21 AND 23;
```

Output Table

Student id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

Q5 list all students in the IT or HR department who are older than 21

Query:

```
SELECT student_id, name, age, department
FROM students
WHERE department
IN ('IT', 'HR')
AND age > 21;
```

Output table:

student_id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

Q6 Show total Credits per department, only for department with more than 5 total credits

Query:

```
SELECT department
SUM (credits) AS Total_credits
FROM courses
Group by department
Having sum (credits) > 5;
```

Output table

department	total_credits
IT	11

Q9 Get the maximum, minimum, and average grade across all enrollments

Query:

```
SELECT MAX (grade)
```

```
AS max-grade,
```

```
MIN (grade)
```

```
AS min-grade,
```

```
Avg (grade)
```

```
AS avg-grade
```

```
FROM enrollments;
```

Output Table

max-grade	min-grade	avg-grade
90	78	84.6

Q10 Count how many enrollments exist per-course

Query:

```
SELECT course-id,
```

```
COUNT (*)
```

```
AS enrollment-count
```

```
FROM enrollment
```

```
GROUP BY course-id;
```

Output Table

course-id	enrollment-count
101	2
102	1
104	1
105	1

① Find total salary and total bonus per department

Query:

```
SELECT department,  
SUM (salary)  
AS total_salary,  
SUM (bonus)  
AS total_bonus  
FROM salaries  
GROUP BY department;
```

Output table

department	total salary	total bonus
IT	122000	10500
HR	109000	7500
Finance	70000	6000

② Show departments where average salary is above 55,000

Query:

```
SELECT department,  
AVG (salary)  
AS avg_salary  
FROM salaries  
GROUP BY department  
HAVING AVG (salary) > 55000;
```

Output table

department	avg_salary
IT	61000
Finance	70000

(13) List employees whose salary plus bonus is greater than 60,000

Query:

```
SELECT employee_id, name, salary, bonus, (salary + bonus)
AS total_compensation
FROM Salaries
WHERE (salary + bonus) > 60000;
```

Output table

employee_id	name	salary	bonus	total_compensation
1	Tom	60000	5000	65000
3	Spike	70000	6000	76000
4	Tyke	62000	5500	67500

(14) Show total and average budget per department..only include departments with average budget above 70000

Query:

```
SELECT department
SUM (budget) AS total_budget,
AVG avg_budget
FROM Projects
GROUP BY department
HAVING AVG (budget) > 70000;
```

Output table

department	total_budget	avg_budget
IT	270000	135000

15) List all projects with budgets between 50000 & 120000 excluding the marketing department

Query:

```
SELECT project_id, project_name, department, budget
FROM projects
WHERE Budget
BETWEEN 50000 AND 120000
AND department != 'Marketing';
```

Output Table

Project_id	Project_name	department	budget
1	AI-App	IT	120000
2	Payroll System	Finance	80000
5	HR Portal	HR	50000